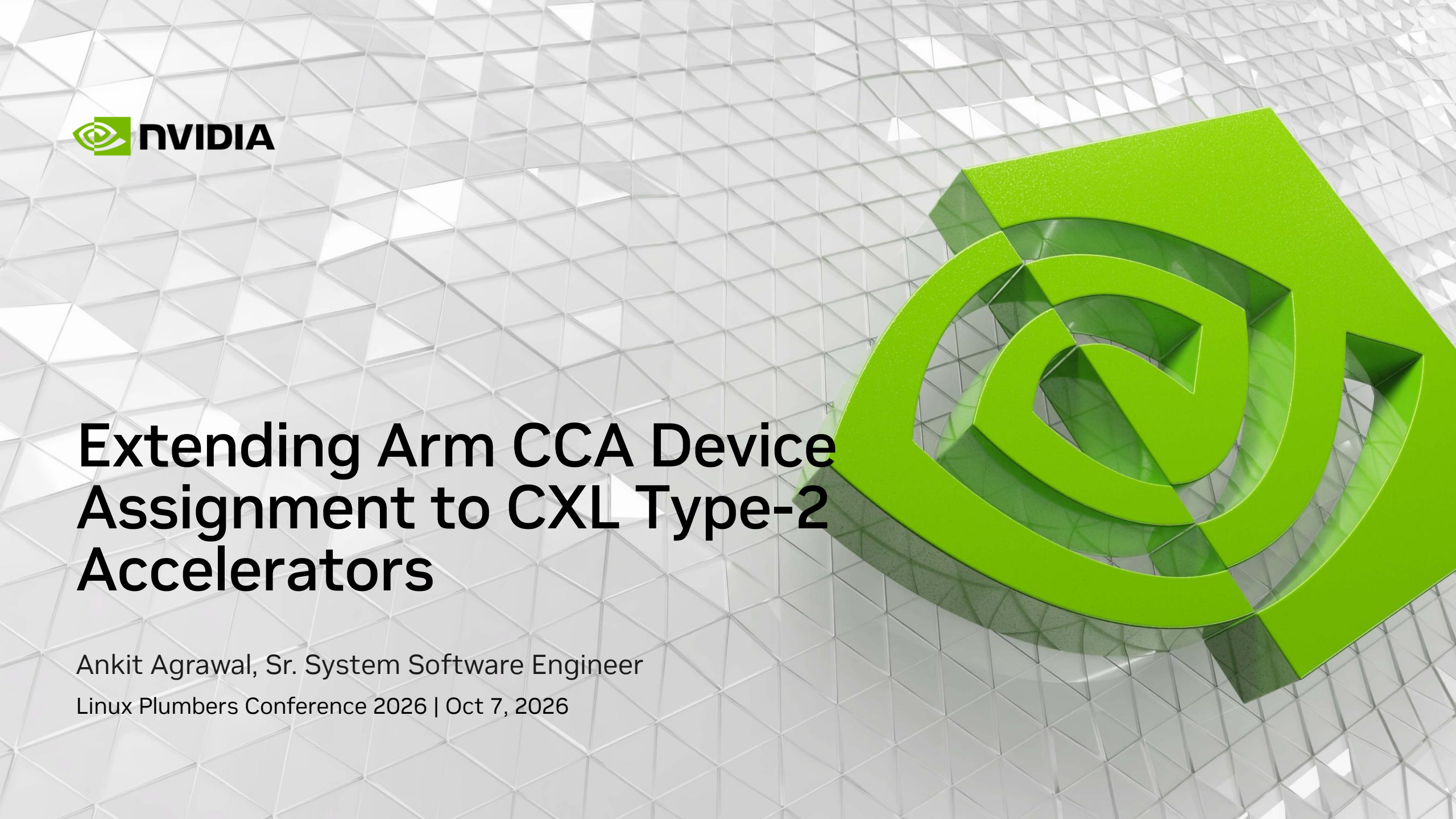


Extending Arm CCA Device Assignment to CXL Type-2 Accelerators

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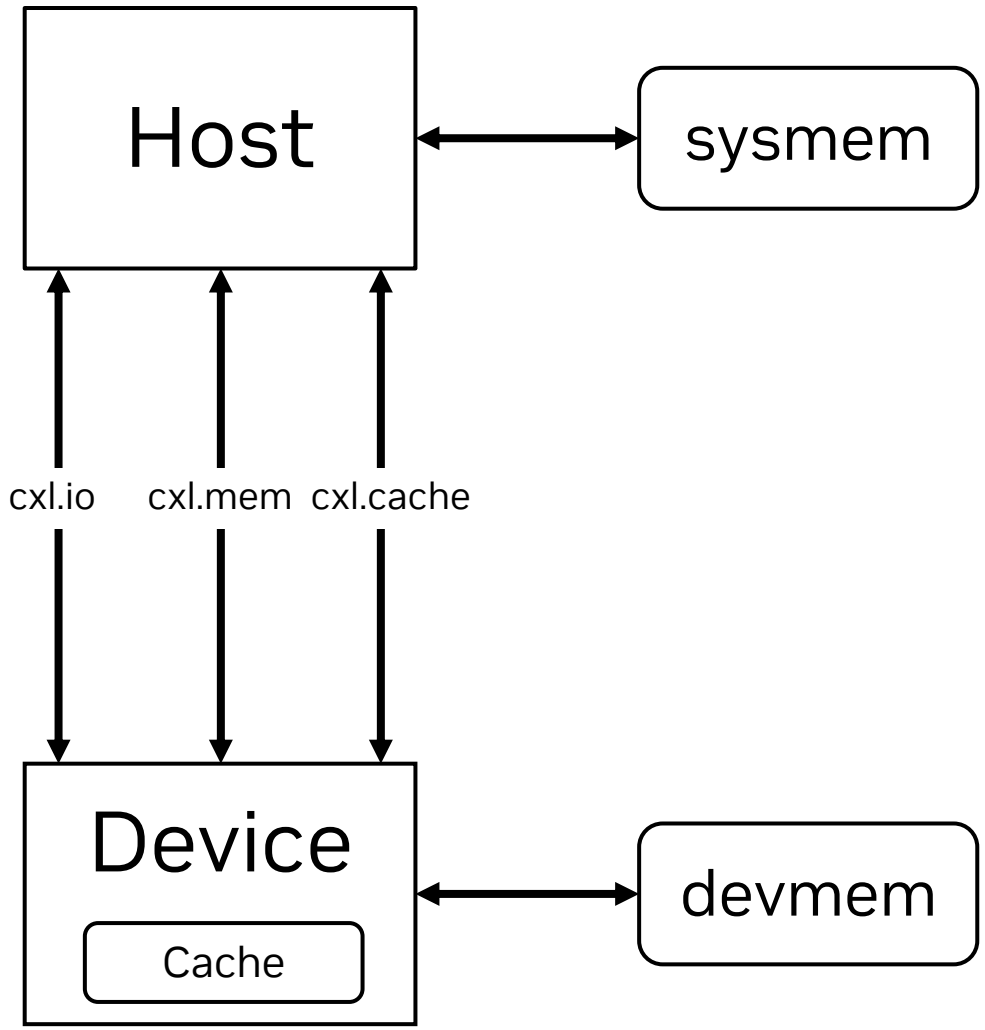
Background

CXL

- Compute Express Link is an open, consortium-backed interconnect for coherent connectivity between CPUs, accelerators and memory.
- Enables the CPU and device to share a coherent view of memory.
- It has three sub-protocols:
  - CXL.io is PCIe-style config, MMIO and DMA.
  - CXL.cache lets a device coherently cache host memory.
  - CXL.mem lets the host load/store device-attached memory coherently.

Type-2 Device

- CXL Type-2 device (such as accelerators) supports all the three sub-protocols.
  - Device memory appears to the host as ordinary cache-coherent system memory.
  - The device can also cache host sysram coherently
- Software sees multiple discovery sources:
  - DVSECs in config space
  - Component registers in MMIO, reached through the Register Locator DVSEC - includes the HDM decoders.
  - ACPI CEDT/CFMWS, which describes the host physical address (HPA) windows.



HDM Decoders

- HDM decoders (Host-managed Device Memory) map host/VM physical addresses (IPA/HPA) to device physical addresses (DPA).
- Setting the COMMIT bit on a decoder makes it live. A HDM decoder has to be committed with the region information.
- Software can read the committed HDM decoders to get the HPA/IPA range of the CXL region
- A device can have one or more HDM decoders.

- Precommitted
  Firmware Programmed HPA base and size before the OS runs. Range readable from the HDM decoder registers.
- Non Precommitted
  Runtime programmed/committed by the driver when cxl_create_region() is called.

Extend CCA DA to cover CXL Type-2 devices

Upstream has the PCI/TSM core with PCIe IDE and TDISP. CXL-specific changes are not designed.

Design Item	Component
Host TSM determination of CXL region information.	Host Kernel
Connect coherent stream for CXL region (termed as region registration in the deck)	Host Kernel
A CXL window in the guest physical space through CFMWS	QEMU/VMM
HDM decoders presented as committed to the guest.	QEMU/VMM
Guest TSM parsing of the interface report and determining the coherent CXL region.	VM Kernel
Guest TSM determination of CXL range IPA for validation for device accept in the Realm.	VM Kernel

Design Choices - Host



TSM needs CXL range and register with RMM

Background

- RMM v2.0 has the primitive COH to register the coherent range.
- The CXL region is valid only after the HDM decoder commit.
- No coherent CXL region connect/registration mechanism in the kernel.

Requirement

- TSM needs to determine the committed CXL region and register with RMM during connect
- Implement coherent stream registration mechanism.

Proposal

Save the region as resource during enumeration, then connect

- Since firmware set the HPA base and size before the OS boots, region information available during enumeration
- Fetch and track as a device resource similar to BAR.
- Introduce new registration API for coherent regions similar to cca_pdev_create_ncoh_stream().

PROS

- CXL region handling similar to BAR.

CONS

- Additional member in pci dev

Question: Acceptable to have the coh_resource[] in the pci_dev?

Proposed Sequence – Precommitted Decoders

1	HDM decoder is programmed and committed for the device.	FIRMWARE / BIOS
2	PCI enumeration fetches the already committed region's HPA and records it into pdev->coh_resource[]	HOST PCI
3	sysfs: echo 1 > .../tsm/connect	USER → HOST TSM
4	cca_pdev_create() → RMI_PDEV_STREAM_CONNECT(type = COH, NCOH), using HPA from step 2	HOST TSM → RMM
5	rmi_stream_handle_coh / _ncoh	RMM → HOST TSM
6	sysfs: bind the device to vfio	USER → VFIO-CXL
7	Probe: claim the precommitted coherent region (already committed and validated)	VFIO-CXL

Probe and connect ordering

Specific to Non Precommitted decoders

Background

- The HDM decoder programming and the CXL region creation happen during vfio probe.
- Device BAR regions are registered during TSM connect on the host.
- Implicit ordering present with vfio probe -> connect.

PRE-COMMITTED ■ **Nothing to solve** — Range programmed by the firmware and exists before either probe or connect.

Proposal 1: Defer the region registration

- Do the CXL region registration with RMM in vfio probe after decoder commit, if region available at connect.

PROS: Supports any probe <-> connect order.

CONS: Different registration timing v/s BAR (at connect)

Proposal 2: Enforce the sequence

- Make the ordering implicit by rejecting connect before probe.
- Read HDM decoder at registration. Fail if HDM uncommitted.

PROS: CXL region registration occur with the BARs at connect

CONS: Hard ordering requirement may not be acceptable for some use cases.

Question: Is there a probe <-> connect ordering requirement?

Design Choices - VM



Circular dependency between validation and driver probe

Specific to Non Precommitted decoders

Background

- CXL region valid after HDM decoder commit.
- TSM coherent region validation needs a committed decoder for acceptance.
- Probe cannot occur until device region already validated and trust granted.

Proposal

Restrict the VM to firmware precommitted ranges only

- CXL ranges firmware programmed and committed before the OS runs.
- Range existence guaranteed before any validation or driver probe, breaking the circular dependency.

PROS

- No additional code needed besides the VM firmware code to program the HDM decoders.

CONS

- No support for Non Precommitted HDM decoders in the VM

Question: Can we enforce a requirement to have precommitted decoder for CC support?

Only one coherent region identified by the spec

Background

- MMIO regions are identified by BAR index.
- A CXL device can have multiple HDM regions represented by individual HDM decoders.
- Only one coherent region has an identifier in the RME spec.

> Arm Realm Management Extension (RME) System Architecture :
> B2.3.5.2 General coherent device requirements

Requirement

- Add support for multiple CXL regions in a device
- One identifier space, no collision with BAR indices

Proposal

Grow index downward from the top

- Extend the original idea of putting the coherent region index at the top.
- 0-5 BAR Indices – Untouched
- ~0-~n Coherent Region Indices - New

PROS

- Simple convention

CONS

- Indexing proposal has no backing in the spec.

Question: is a Linux-side convention enough, or does this belong in the spec?

Proposed Sequence – Precommitted Decoders

1	HDM decoder is programmed and committed for the VM's view of the device	FIRMWARE / BIOS
2	PCI enumeration fetches the already committed region's IPA and records it into pdev->coh_resource[]	GUEST PCI
3	sysfs: Move to run state	USER → GUEST TSM
4	Parse the interface report to get the CXL range using IPA from step 2	GUEST TSM
5	TDISP RUN — the coherent window can be treated as private	GUEST TSM
6	Probe: claim the precommitted coherent region (already committed and validated)	NVIDIA DRIVER

Summary

01 TSM needs CXL range and register with RMM

Host | Precommitted · Non Precommitted

Acceptable to have the coh_resource[] in the pci_dev?

02 Probe and stream connect ordering

Host | Non Precommitted

Is there a probe <-> connect ordering requirement?

03 Circular dependency between validation and driver probe

VM | Non Precommitted

Can we enforce a requirement to have precommitted decoder in the VM for Confidential Computing support?

04 Only one coherent region in spec

VM | Precommitted · Non Precommitted

Is ~0, ~1, ... the right identifier scheme — and is a Linux convention enough or does it also need work in the spec?

Relevant Patches

CCA support for type-2 CXL device - VM patch series
[RFC PATCH 0/4] PCI/TSM: Resolve TDISP coherent (CXL) ranges from precommitted HDM decoders

CXL Type-2 device reset support
[PATCH v14 00/16] PCI/CXL: Add CXL reset support for Type 2 devices - Srirangan Madhavan

VFIO-CXL support
[PATCH v5 00/27] vfio/pci: Add CXL Type-2 device passthrough support

CXL Type-2 device passthrough - Qemu
[PATCH v2 00/10] QEMU: CXL Type-2 device passthrough via vfio-pci

References

CXL Specification Rev 4.0, Ver 1.0 (CXL Consortium)
https://computeexpresslink.org/cxl-specification/

Arm RME / RMM Specification 2.0 (DEN0137, 2.0-bet3)
https://support.arm.com/documentation/den0137/2-0bet3

THANK YOU

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